MaoLin He

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https://github.com/rinmy23

Educations

The University of Melbourne QS Top50

Master of Information Technology (with Distinction)

- Related Subjects: Natural Language Processing (85), Declarative Programming (85), Distributed Systems (85), Advanced Database Systems(82), Research Methods (82), Research Project (81), Advanced Studies in Computing (81)
- GPA: 3.89/4.0 First Class Honours
- Supervisor: A/Prof Brian Chapman

Beijing Normal University 985 211 Double 1st-Class

Bachelor of Science (Computer Science and Technology)

- Average Score of professional courses: 82/100
- Bronze Medal of China Collegiate Programming Contest (Twice)

Research Interest

My research interests focus on trustworthy large language models, particularly how they support reliable interaction in complex scenario. Specifically, it includes: Reliable knowledge retrieval and representation, Faithful reasoning and grounding in LLMs, Interpretability and robustness of model outputs.

Research Experiences

Towards semantic reliable clinical QA: Query pipeline optimization for cancer patient question answering systems Advisor: A/Prof Brian Chapman

- Information Retrieval Optimization: Developed **Clinical Hybrid Semantic-Symbolic Document Retrieval (CHSDR),** an innovative method that includes: Boolean-constrained real-time search with adaptive query execution, Semantic–symbolic retrieval fusion with metadata-aware document control.
- Chunk Optimization: Introduced **Semantic Enhanced Overlap Segmentation (SEOS)**, a novel text segmentation approach that combines sentence semantics with embedding model influences while leveraging chunk overlap for enriched context.
- This study, by combining the two optimizations with LlamaIndex, improved the response accuracy of the Claude-3-haiku model by **5%**. This work is now submitted to **ACL**.

Creating a Cancer Patient-facing QA System using LLMs Advisor: A/Prof Brian Chapman

- Developed **Hierarchical Evidence-Processing (Hier-EP)** which uses multi-factor filtering and adaptive granularity control to extract high-quality evidence, improving response accuracy by **8%** in the Claude-3-haiku model.
- Integrated source tracking, citations, and readability adjustments to enhance output understandability and verification.
- Proposed the **Multi-Criteria Adaptive RAG (MCA-RAG)**, applying multiple criteria for selecting optimal retrieval approaches. A key component, the **Dual-LLM Collaborative ReAct-based Multi-Step Retrieval**, combines LLaMA-3-8B for stepwise responses with LLaMA-3-70B for high-level reasoning. MCA-RAGenhanced **LLaMA-3-8B's accuracy to match GPT-4**.

Selected Assessment Projects

Automated Fact Checking for Climate Science Claims Feb 2023 - Jun 2023

Individual research project of Natural Language Processing (COMP90042)

- Developed information retrieval (IR) function using Dense Passage Retrieval (DPR) to search for the most related evidence for the given claim in a dataset of over 1208k+ records
- Analyzed data and results, pondered causes and investigated relevant papers; Adjusted hyperparameters and applied novel methods (Error-Prone Negative Sampling and Adaptive Retrieval top-n), improving IR model's F1 score from 0.12 to 0.16
- Implemented a classification function for claim verification using 4 pre-trained models (BERT, ALBERT, Roberta, Distilroberta) and an ensemble approach with the highest accuracy of 0.56

Reversi Autonomous Agent Jul 2022 - Nov 2022

Group research work of AI Planning for Autonomy (COMP90054)

- Implemented 3 different algorithms (BFS, MCT and alpha-beta); Improvements are made to each of these three algorithms based on the rules of the game
- Collaborated with team-mates on developing a Wiki, documenting and describing solutions; Received 31.25/35 for the assignment

Services

Semester 2, 2023 Orientation Volunteering tour guide Jul 2023 - Aug 2023

Showed 25 commencing students around campus during orientation week; Introduced the layout of University and Faculty facilities

Additional Informations

- **ProgrammingLanguages:** C/C++,Java,Python, R
- **DeepLearningFramework:** PyTorch,Keras
- **Tools:** LATEX, Llamaindex
- **Languages**: Chinese (native), English (PTE: 68)